

India EV Aftersales Strategy

A Simulation-Based Strategic Analysis of OEM Service Networks

₹5,443 Cr

Aftersales TAM by FY30

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Delhi Technological University | 2027

Python · Power BI · Monte Carlo · Pandas

Methodology Note: All figures are derived from a 67,000+ record simulated dataset, calibrated to public benchmarks (VAHAN, SMEV 2023-24). This is a strategic modeling exercise, not an audit of real OEM financials.

India's EV service infrastructure is not built for what is coming

₹5,443 Cr

Market by FY30

EV aftersales TAM reaches ~Rs 5,443 Cr by FY30 in this simulation, based on projected fleet growth and Rs 8,500/vehicle/year aftersales spend.

74.3%

Centers below breakeven

In the simulated network, 74.3% of Tier-3 service centers operate below breakeven every month due to low throughput vs fixed costs.

32.6%

Claims are OTA-avoidable

32.6% of simulated warranty claims (36.9% of warranty spend) could be resolved remotely via software updates instead of a workshop visit.

This study models 3 strategic levers to address service network economics and warranty cost efficiency

Hypothesis-driven framework across 3 MECE value levers

Can India EV OEMs improve aftersales EBITDA economics through targeted operational interventions?

Parts Pricing

- H1a: EV parts underpriced vs ICE equivalents
- H1b: Battery module pricing leaves margin on the table
- H1c: Aftermarket cannibalization reduces OEM parts revenue

Service Network ROI

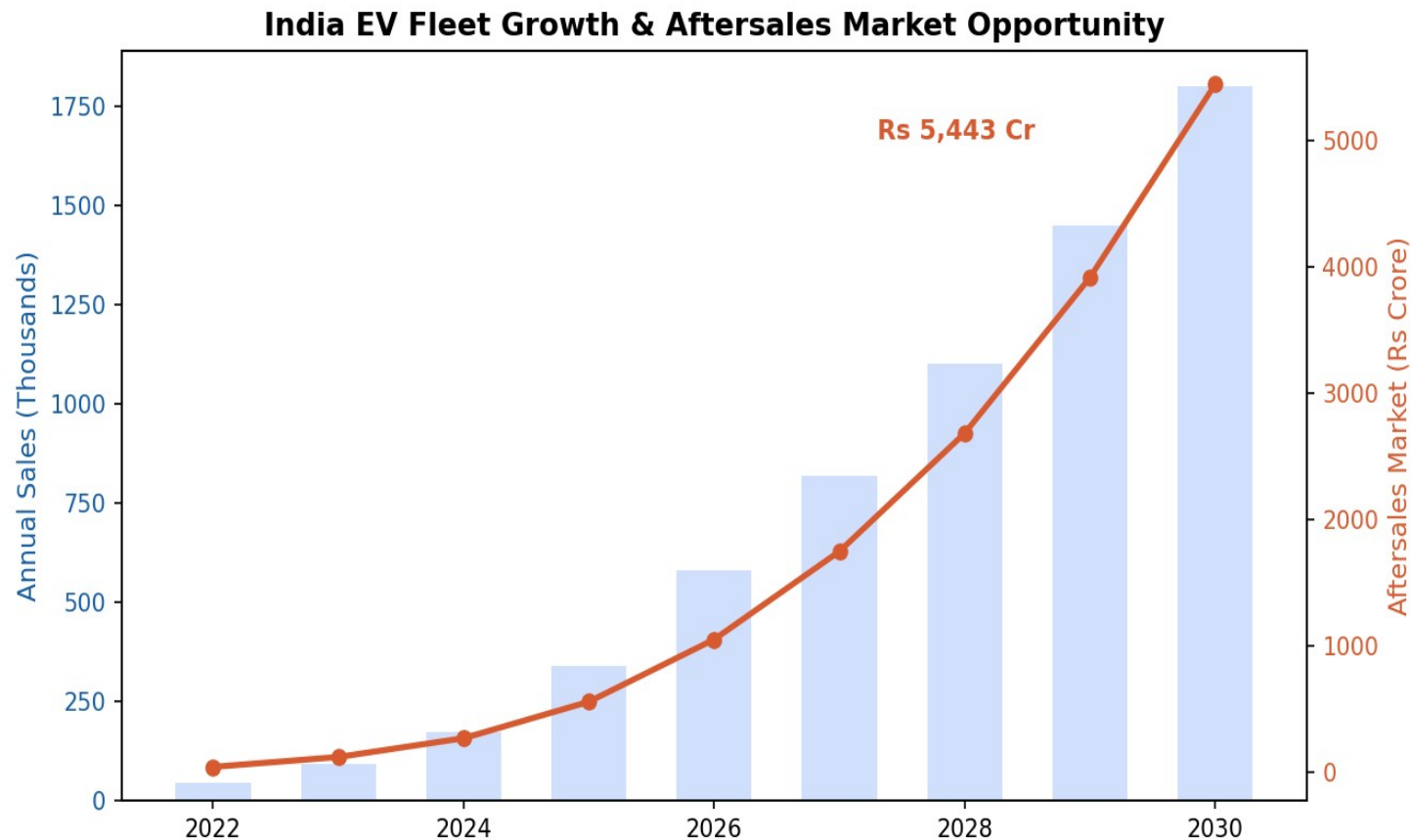
- H2a: Tier-3 dealer throughput below breakeven threshold
- H2b: Mobile vans have lower fixed cost vs fixed centers
- H2c: OTA updates reduce avoidable workshop visits

Warranty Cost

- H3a: Reserve levels can be right-sized using simulation
- H3b: Telematics/OTA can reduce avoidable claims
- H3c: Climate zone affects battery failure patterns

FINDING 1

Simulated EV aftersales TAM reaches ~Rs 5,443 Cr by FY30



Rs 5,443 Cr

Aftersales TAM, FY30

64L+

Cumulative EV fleet, FY30

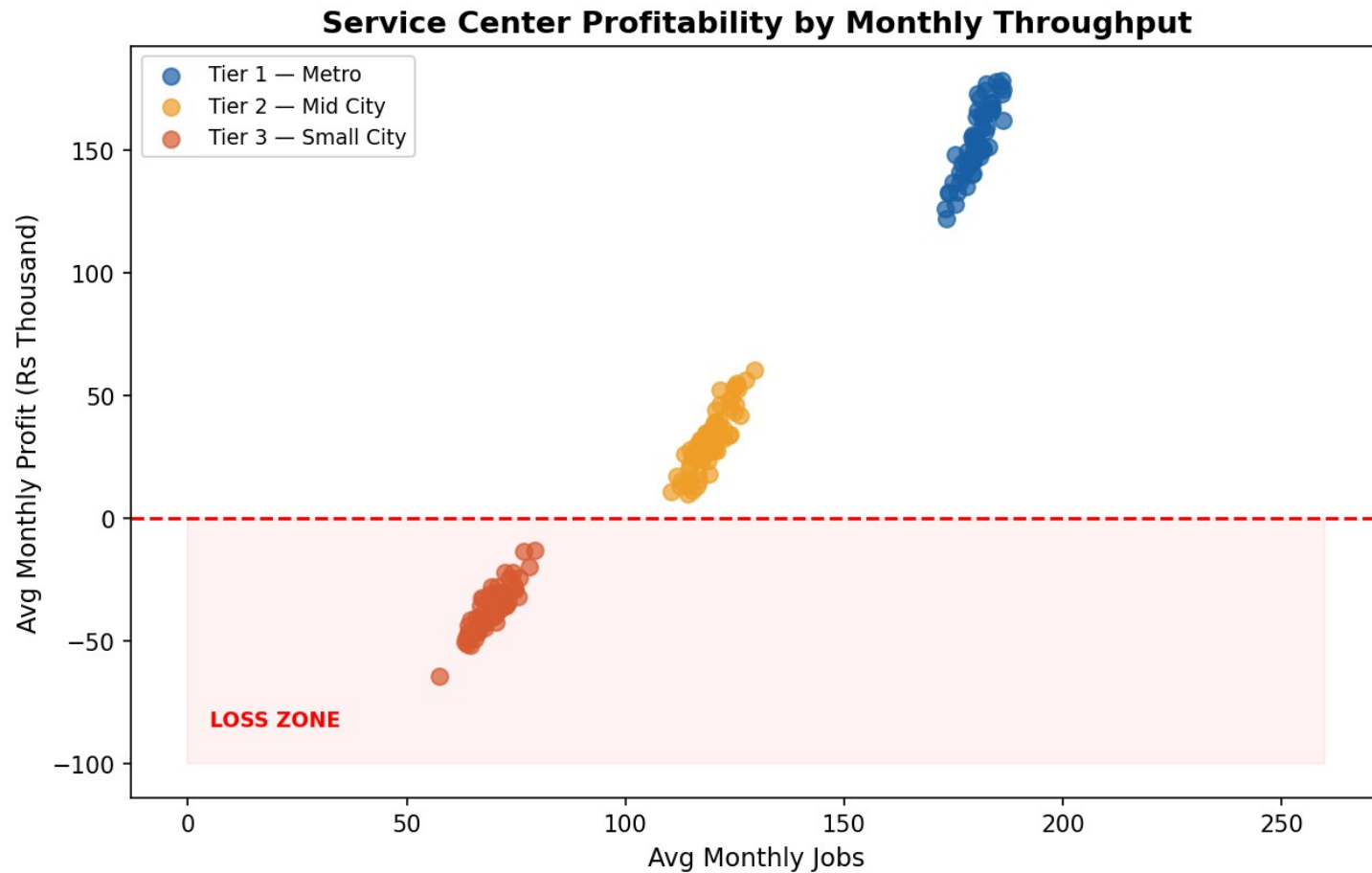
Rs 8,500

Avg spend / vehicle / year

Tata Motors holds 53.1% fleet share in this dataset — the single largest stake in the modeled opportunity.

FINDING 2

74.3% of Tier-3 service centers operate below breakeven



Tier 1 — Metro

180.2 actual | 134.1 breakeven

+Rs 1.54L/mo

Tier 2 — Mid City

119.3 actual | 107.7 breakeven

+Rs 32K/mo

Tier 3 — Small City

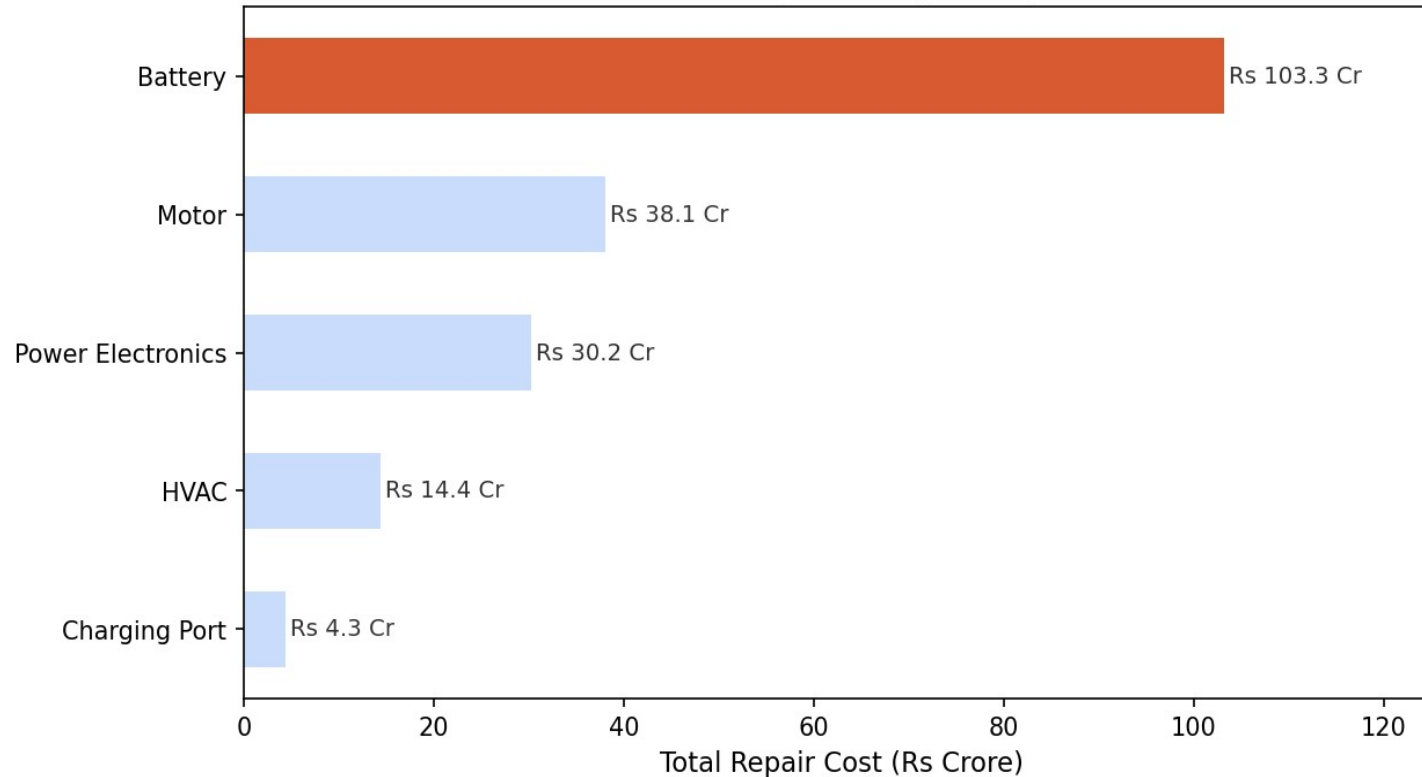
69.5 actual | 88.3 breakeven

-Rs 36.5K/mo

FINDING 3

Battery accounts for 54.3% of total simulated warranty spend

Total Warranty Cost by Component



Rs 103.3 Cr

Battery total warranty cost

Rs 59,474

Avg battery repair cost

54.3%

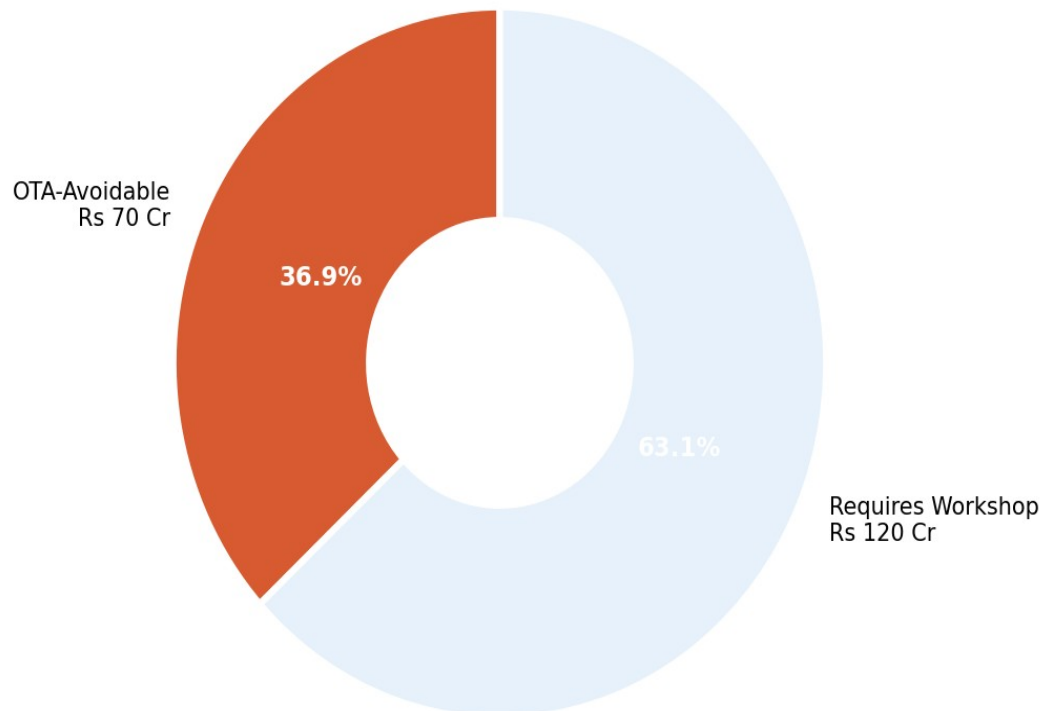
Battery share of warranty spend

Climate insight: Hot-Humid zones show a claim-rate index of 1.61 vs 1.00 for Temperate zones — directionally consistent with known battery degradation under heat.

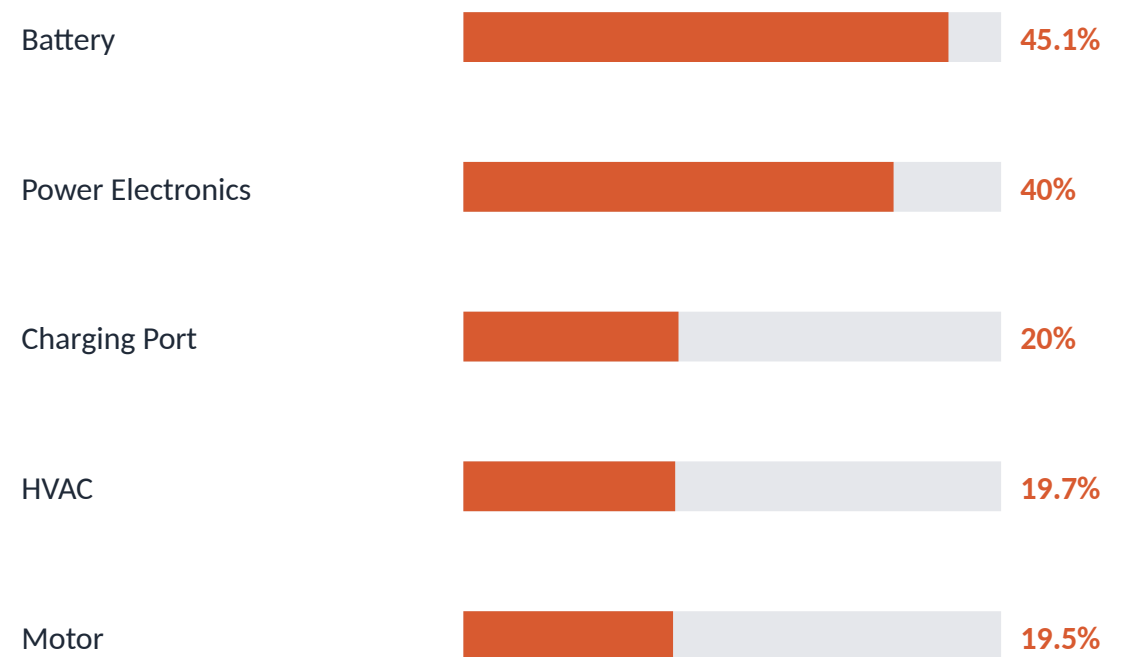
FINDING 4

36.9% of warranty spend is OTA-avoidable (32.6% of claims)

Warranty Spend: OTA-Avoidable vs Workshop Required

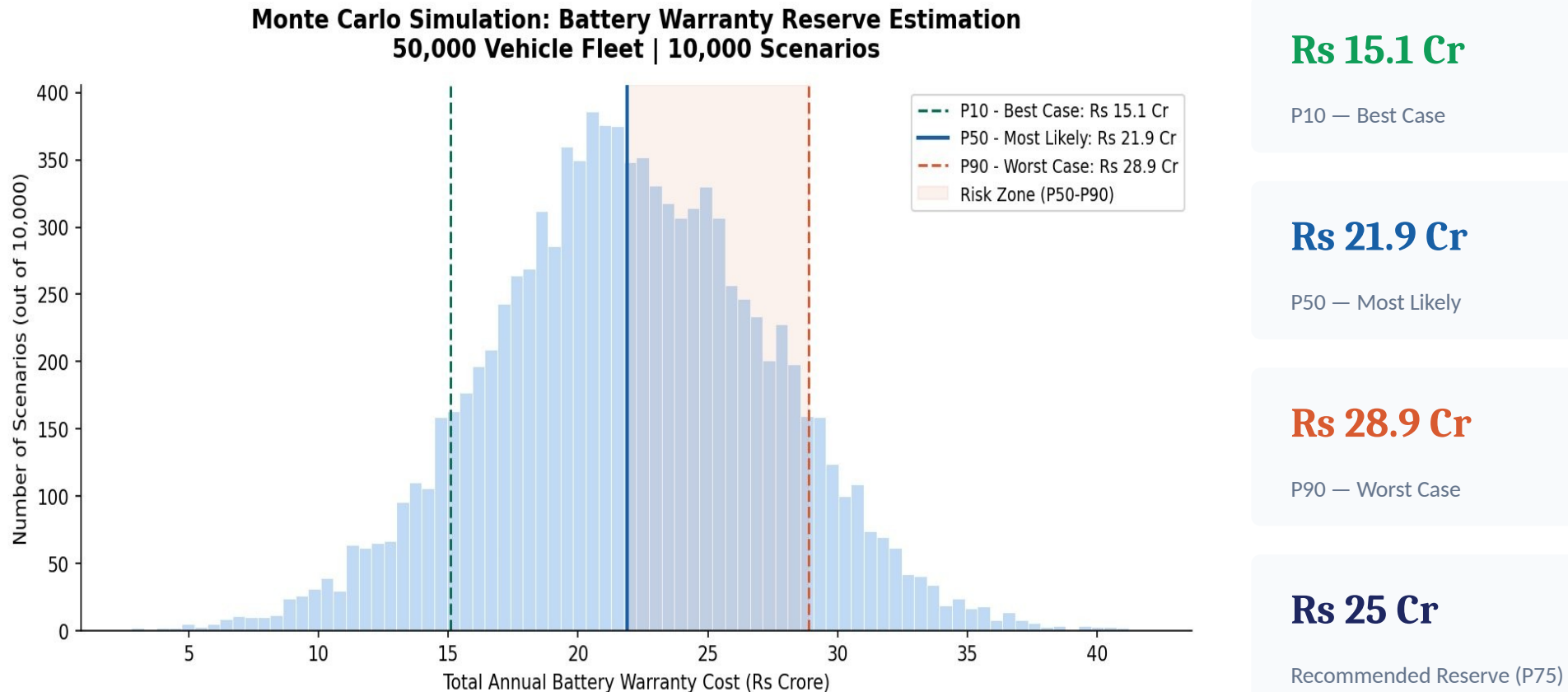


OTA-avoidable rate by component



Tesla benchmark: 35-40% of service visits resolved via OTA. India OEMs today: ~5%.

10,000-run simulation estimates the real warranty reserve range



Rs 15.1 Cr

P10 — Best Case

Rs 21.9 Cr

P50 — Most Likely

Rs 28.9 Cr

P90 — Worst Case

Rs 25 Cr

Recommended Reserve (P75)

Methodology: 10,000 simulations | Weibull failure distribution (shape 1.5, scale 60) | Lognormal repair cost (mean Rs 55K, std Rs 15K) | 50,000-vehicle fleet | Climate multipliers applied

Three strategic levers to improve aftersales economics

01

Parts Pricing Optimization

Review pricing on high-velocity SKUs relative to ICE benchmarks. Current model shows EV parts gross margin (28-32%) below ICE benchmark (35-40%), suggesting room to reprice without major volume loss.

DO NOW • 0-6 months

02

OTA Software Deflection Program

Deploy proactive telematics-triggered OTA updates, prioritizing Battery (45.1% avoidable) and Power Electronics (40.0% avoidable) — the two components with the highest avoidable-claim rates.

PLAN • 6-18 months

03

Mobile Service Vans — Tier 3

Replace fixed Tier-3 service centers with mobile vans to reduce fixed costs (modeled: Rs 1.8L to Rs 60K/month), closing the breakeven gap (69.5 actual vs 88.3 required jobs/month).

PLAN • 12-24 months

Note: Directional, modeled impact only. No aggregate Rs Cr figure is claimed without a fully itemized, bottom-up calculation per lever.